



Dell Digital Cities

# Artificial Intelligence and Gen AI in Smart Cities

By John Lockhart  
Edge Product Management, Global Digital Cities



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# Introduction

Artificial Intelligence (AI) delivers cities a wide array of capabilities in handling multitudes of data bring to bear essential elements to assist in processing, categorizing, and determining patterns in the myriad of a city's many segments. Understanding the nuances of the data patterns allows cities, leaders, and administrators to improve community services with the limited resources at their disposal. One of the tools in particular, Generative AI (Gen AI), is moving the needle in ways once deemed impossible or improbable, delivering much needed automation to areas where budgets are constrained and cities are working to do more with less. As this document assesses impacts and ways AI and Gen AI will be used to enhance, optimize, and predict patterns in people, events, traffic, and many further use cases, the usage of AI within city operation centers from a departmental level through integration of entire city systems into a holistic framework of data sets will make the role of city leaders and master planners more simplified in the future.

Since the advent of collaborative artificial intelligence engines to the current state of the newest and most innovative tools, Gen AI brings a vast array of potential options for government to provide services to communities and citizens alike. AI and Gen AI's capabilities will help cities to simplify, optimize, and automate city services including urban modeling, master planning, and efficient sustainable development of a city's evolving needs. Gen AI technology provides city leaders and administrators efficient and accurate insights to the pulse of a city, rapidly adapting to the changing environment and improving citizen livability.



The initial critical role played by Gen AI would be in creating efficiencies for a wide range of understaffed city services without the need to increase resources, accurately responding to a city's needs. Where in the past, this required administrators to be versed in all aspects of city obligations, from policies to action to response, Gen AI offers the capability to have chatbots fill the gap, effectively reducing the human demand to support citizens' concerns. Gen AI further evaluates normally disparate data sets including both historical and real-time data trends to predict, optimize and automate labor-intensive administrative functions.

The Dell Digital Cities Solutions Team focuses on creating a tapestry of horizontally interconnected systems and subsystems forming a data oasis consisting of a multitude of data lakes, data lake houses, and other stored data sets to correlate data insights, driving a city's strategic vision by delivering meaningful outcomes. These outcomes are accelerated by five key pillars: Integrated Operations, Sustainable City Development, Urban Mobility, Public Safety, and eGovernment Services.